

I2200 Assignment 1

Note: Please provide a well-documented copy of your code (MATLAB or any other language of your choice) for the questions that ask you to perform operations on images.

Question 1) [Sampling/Reconstruction – 40 pts.]

1-1) Down-sample image “lenna.jpg” of size 256x256 to represent it with size of 64x64.

1-2) Reconstruct the image at original size of 256x256 by interpolation from the 64x64 samples using the following schemes. In both cases make sure you include the resulting images in your response.

- (a) Use the Nearest-Neighbor (Zero-order) interpolation function
- (b) Use the bilinear interpolation function

Compute the average error for both cases, compare them, and comment on your results.

Note: *You are required and will be graded for your own implementation of the down-sample, reconstruction, and interpolation.* However, you can use MATLAB’s built in functions to check your results.

Question 2) [Quantization – 20 pts.]

Again take image “lenna.jpg” of size 256x256, which is originally quantized with 8 bits per pixel. Quantize the range of gray-level values of the pixels in this image with only B=4 bits. Use the linear quantizer (uniform optimal quantizer) to perform the quantization. Include the resulting image quantized with B=4 bits in your response.

Compute the average error between the result images. Comment on the subjective quality of the images in both cases.

Question 3) [Color – 40 pts.] – [Do this after Lecture 3.](#)

3-1) Covert image “lenna_color.tif” to gray image by using the following equation:

$$\text{Intensity} = 0.2989 \cdot \text{red} + 0.5870 \cdot \text{green} + 0.1140 \cdot \text{blue}$$

Then compare your result image with the result image generated by Matlab built-in function.

3-2) Get the image from each channel of red, green, and blue respectively and compare each channel with the gray image generated in 3-1).